

Project Statement: Cafe Management System

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PROJECT	CAFE MANAGEMENT
PROJECT TYPE	Python Console Application

1. Problem Statement

Managing basic cafe operations manually can be inconvenient and time-consuming, especially when menu information, customer orders, quantities, and bill calculations have to be handled repeatedly. Manual calculation can also lead to simple mistakes in item amounts or the final total.

To address this, the Cafe Management System provides a simple Python-based command-line application. It allows a user to view cafe information and menu items, place an order by selecting items and quantities, and generate a bill automatically from the stored order.

The system is designed as a small academic project to demonstrate how basic Python programming concepts can be applied to a practical cafe-management problem.

2. Scope of the Project

The project is an in-memory cafe management application designed to demonstrate structured programming using Python.

In-Scope:

- Displaying cafe details such as cafe name, location and operating timings.
- Displaying the available menu items and their prices.
- Menu-driven command-line interaction.
- Accepting customer orders and quantities during runtime.
- Validating menu item names and rejecting invalid quantities.
- Using a list of tuples to store the current order.
- Calculating the amount for each ordered item and the final bill.
- Using functions, dictionaries, lists, tuples, loops, conditional statements and try-except handling.

Out-of-Scope / Future Enhancements:

- Permanent data storage using files or a database.
- Graphical user interface or web interface.
- Online payment or payment gateway integration.
- Customer login or multi-role authentication.
- Advanced inventory, billing or sales-reporting features.

3. Target Users

- Cafe Staff / Operators: Users who need a simple way to view the menu, record an order and calculate the bill.
- Cafe Owner: A basic local tool for demonstrating the menu and order process in a small cafe environment.

- Programming Students & Educators: Users who can study the project as a practical example of Python dictionaries, lists, tuples, functions, loops and control structures.

4. High-Level Features

- 1. Cafe Information:** Displays the cafe name as Royal Cafe, location as Sehore, and operating timings as 10:00 AM - 10:00 PM.
- 2. Menu Display:** Displays Coffee – Rs. 80, Tea – Rs. 40, Burger – Rs. 120, Pizza – Rs. 200, and Sandwich – Rs. 100.
- 3. Place Order:** Allows the user to select a menu item, enter its quantity, and add the item and quantity to the current order. Ordering continues until **done** is entered.
- 4. Input Validation:** Checks whether the entered item exists in the menu and whether the quantity is valid. Quantities less than or equal to zero are rejected, and invalid numeric input is handled using try-except.
- 5. Bill Generation:** Reads the stored order, obtains each item's price from the menu dictionary, multiplies price by quantity, displays each amount and calculates the final bill.
- 6. Interactive Main Menu:** Provides navigation through Cafe Information, Show Menu, Place Order, Generate Bill and Exit.

5. Data Structures Used

Data Structure	Purpose in the Project
Dictionary	Stores menu items as keys and their prices as values.
List	Stores the customer's order entries during program execution.
Tuple	Stores fixed cafe information: name, location and timings.
Tuple in List	Stores each order entry as (item, quantity).

6. Main Functions

- **show_cafe_info()** – Displays the cafe name, location and timings.
- **show_menu()** – Displays all available menu items and prices.
- **take_order()** – Accepts orders, validates quantities and stores item-quantity entries.
- **generate_bill()** – Calculates and displays item-wise amounts and the final bill.
- **main()** – Controls the complete menu-driven application and navigation.

7. Program Flow

The application follows a simple sequence: the user starts the program → the main menu is displayed → the user selects an option → the corresponding function performs the operation → the result is displayed → the program returns to the main menu until Exit is selected.

The current implementation does not include a database, GUI, online service, payment system or permanent file storage. These are outside the scope of the present project.

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